Homework 3

FIN2020 Fall 2023

1. Under certainty, any increasing monotone transformation of a utility function is also a utility function representing the same preferences. Under uncertainty, we must restrict this statement to linear transformations if we are to keep the same preference representation. Give a mathematical as well as an economic interpretation for this.

   Check it with this example. Assume an initial utility function attributes the following values to three perspectives: $u(B) = 100$; $u(M) = 10$; $u(P) = 50$.

   a. Check that with this initial utility function, the lottery $L = (B, M, 0.50) \succ P$.

   b. The proposed transformations are $f(x) = a + b\,x$, $a \geq 0$, $b > 0$ and $g(x) = \ln(x)$. Check that under f, $L \succ P$, but that under g, $P \succ L$.

2. Discuss the equivalence between (x, z, π) and $(x, y, \pi + (1 - \pi)\,\tau)$ when $z = (x, y, \tau)$. Can you think of circumstances under which they would not be viewed as equal?

3. Expected Utility and Risk Aversion

   Consider an investor with von Neumann-Morgenstern expected utility function

   $$U(x, y, \pi) = \pi u(W_0 + x) + (1 - \pi)u(W_0 + y),$$

   defined over lotteries that offer payoff x with probability π and payoff y with probability $1 - \pi$, where W_0 measures the investors wealth before accepting the lottery. As we discussed in class, concavity of the Bernoulli utility function u captures the investor's aversion to risk.

   To verify this with an example, suppose that the Bernoulli utility function takes the specific form

   $$u(c) = \frac{c^{1-\gamma}}{1-\gamma}$$

   so that

   $$u'(c) = c^{-\gamma}$$

   and

   $$u''(c) = -\gamma c^{-\gamma-1}.$$

This last expression for $u''(c)$ shows that the investor will be risk averse if the preference parameter γ is positive and, will become in some sense, "more risk averse" the larger is γ.

Suppose that $W_0 = 10$, so that the investor starts out with \$10, and consider the following three lotteries:

1. Lottery one has $(x, y, \pi) = (5, 0, 1/2)$, so that the investor wins \$5 with probability $1/2$ and gets nothing with probability $1/2$.
2. Lottery two has $(x, y, \pi) = (2.5, 0, 1)$, so that the investor gets \$2.50 with probability one, that is, for sure.
3. Lottery three has $(x, y, \pi) = (2, 0, 1)$, so that the investor gets \$2 with probability one, that is, for sure.

Using these numbers, compute and compare the values of

$$U(x, y, \pi) = \pi u(W_0 + x) + (1 - \pi)u(W_0 + y) = \pi \left[\frac{(W_0 + x)^{1-\gamma}}{1 - \gamma}\right] + (1 - \pi)\left[\frac{(W_0 + y)^{1-\gamma}}{1 - \gamma}\right]$$

for all three lotteries when the investor has $\gamma = 1/2$. Then re-do the exercise with $\gamma = 2$ and $\gamma = 3$. Looking across the results for all values of γ, does the investor ever prefer the risky bet of \$5 versus 0 to the safe option that offers the average of \$2.50 for sure? Why or why not? Then compare the risky bet of \$5 versus 0 to the safe option that offers only \$2 for sure. For which value(s) of γ does the investor prefer the risky bet to the safe choice of \$2 for sure? For which value(s) of γ does the investor prefer the safe choice of \$2 to the risky bet? For which value(s) of γ is the investor indifferent between the risky bet and \$2 for sure?

4. Stochastic Dominance

Consider the following investments:

Investment 1		Investment 2	
Payoff	Prob.	Payoff	Prob.
1	0.25	3	0.33
7	0.50	5	0.33
12	0.25	8	0.34

Check that neither investment dominates the other on the basis of

- The Mean-Variance criterion
- First Order Stochastic Dominance
- Second Order Stochastic Dominance

How could you rank these investments ?